

Powering AI: Capital, Power Bottlenecks and Mapping Adoption

July 24, 2025

Vishwas Patkar – Head of US Credit Strategy | Strategist

Carolyn Campbell – ABS Strategist | Strategist

Lindsay Tyler – IG TMT Credit | Analyst

Stephen Byrd – Global Head of Thematic and Sustainability Research | Equity Strategist

Jamil Ohayia – Thematic and Sustainability Research | Equity Strategist

Todd Castagno – Head of Global Valuation, Accounting & Tax (GVAT)

David Arcaro – U.S. Power, Utilities, and Clean Tech | Lead Analyst

MORGAN STANLEY & CO. LLC

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

Morgan Stanley

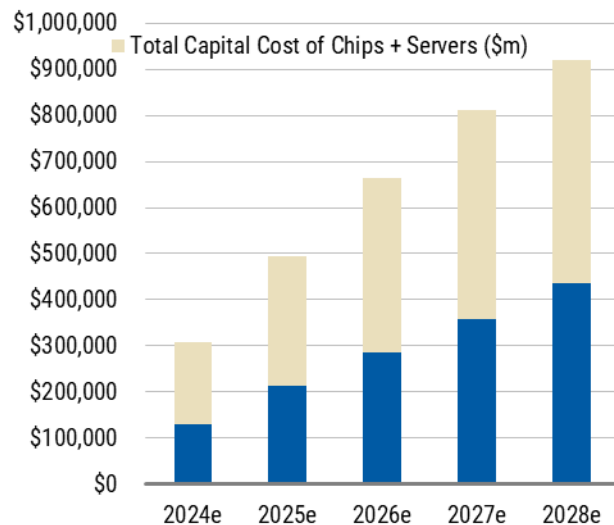
Financing AI Infrastructure

Vishwas Patkar, Carolyn Campbell, Lindsay Tyler

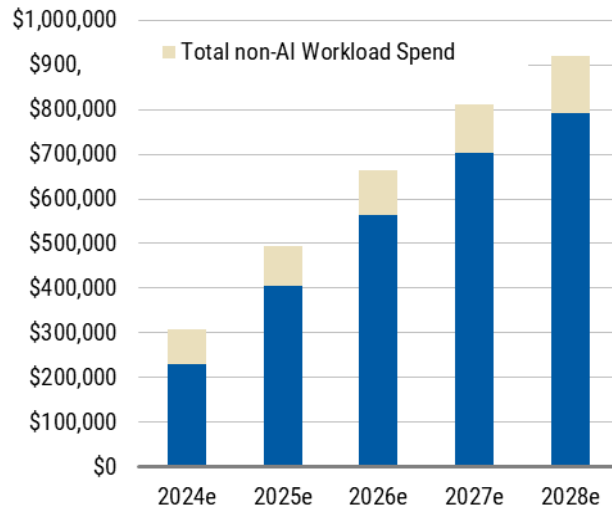
Forecasting ~\$3tr global data center spend through YE28

We estimate \$2.9tr of global data center spend, not inclusive of power investments, through 2028, of which 85% will be for AI-specific data centers

Est. Total Global Spend (\$mm) by Type



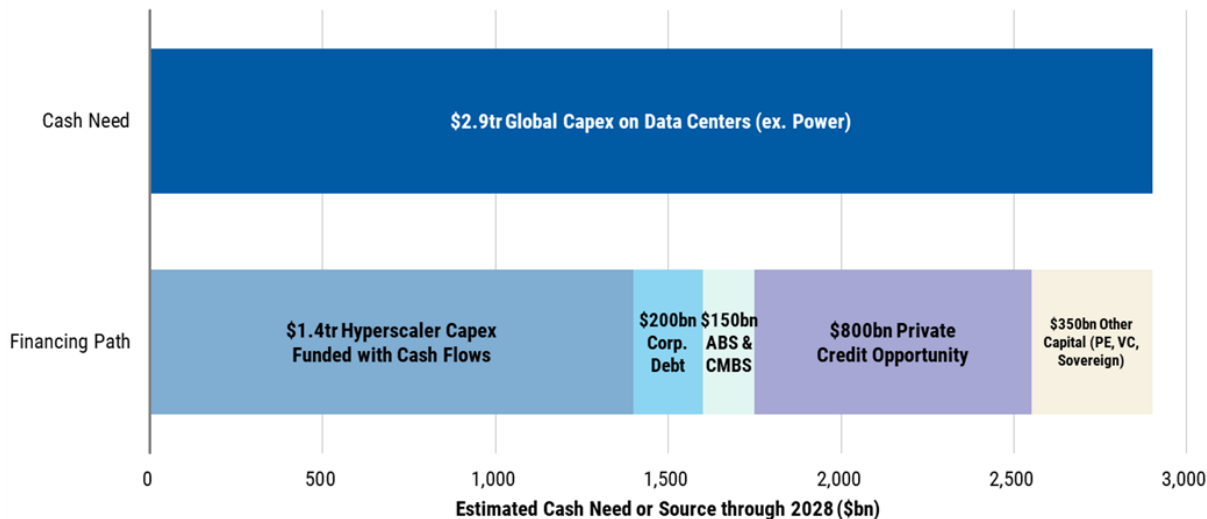
Est. Total Global Spend (\$mm) by Workload



Estimating Incremental Financing Needs and Paths

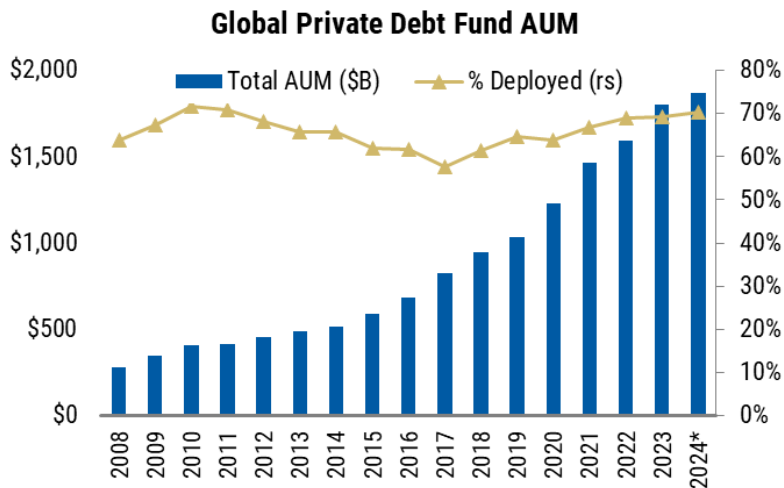
We estimate a \$1.5 trillion gap in data center investment needs that will need to be financed through external markets once we account for hyperscaler capex funded with cash flows.

Estimated financing breakdown for total capex, 2025-28; \$800 billion opportunity for private credit



Credit Markets Have Capacity to Meet Financing Needs Across Private and Public

Credit markets seeing significant inflows, with private credit leading the way



Strong demand-supply dynamics in credit

- Positive real yields have driven inflows from long-term buyers like insurance companies and pension funds.
- New supply/M&A has been muted with the high funding costs, and persistent macro uncertainty.
- Private credit has gained traction as a way to earn excess income, with limited correlation to broader risk assets.
- Asset-based finance, new frontier for insurance balance sheets, providing IG exposure, tailored for DC financing.

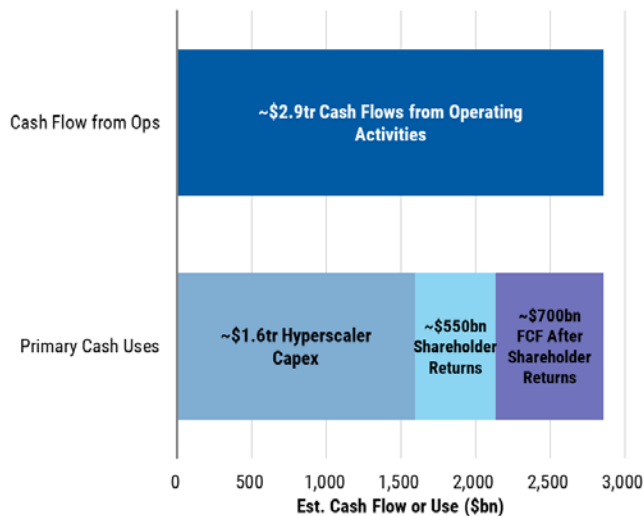
Risks to financing capacity and our estimates?

- Slower growth and sharply lower real yields (beyond Morgan Stanley base case) could be a challenge for continued credit demand.
- Hyperscalers slowing down capex plans could increase reliance on external markets.
- Faster revenue monetization can lead to more issuance in public corporate debt markets.

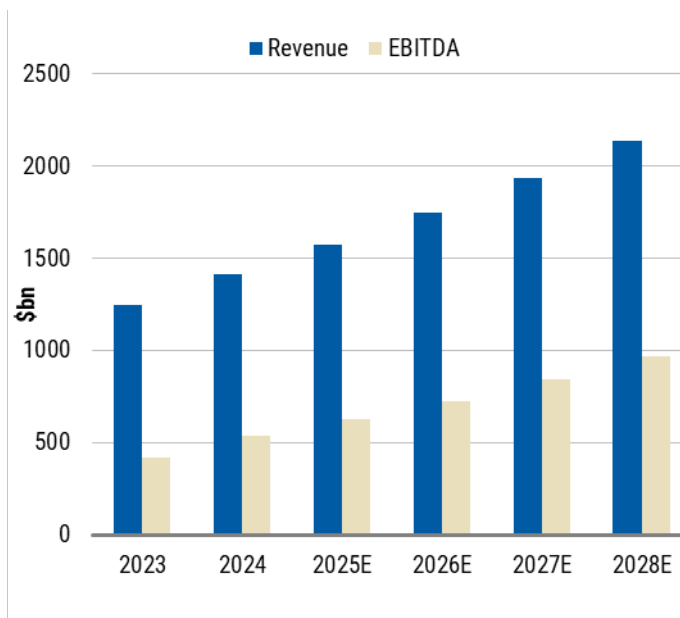
Estimate \$1.4tr hyperscaler capex to be funded with cash flows

Rising cash flows may continue to support growing capex, but shareholder returns, potential acquisitions and other factors will bring practical spend on AI capex down

Hyperscalers' Cash Flows & Uses, Estimates



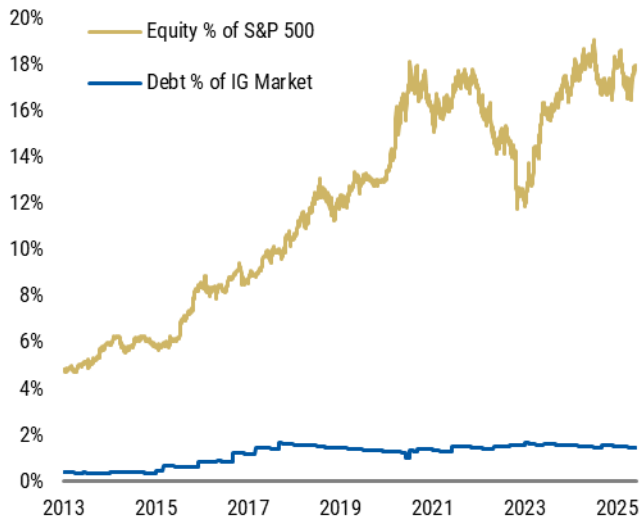
Hyperscaler Revenue & EBITDA Expectations



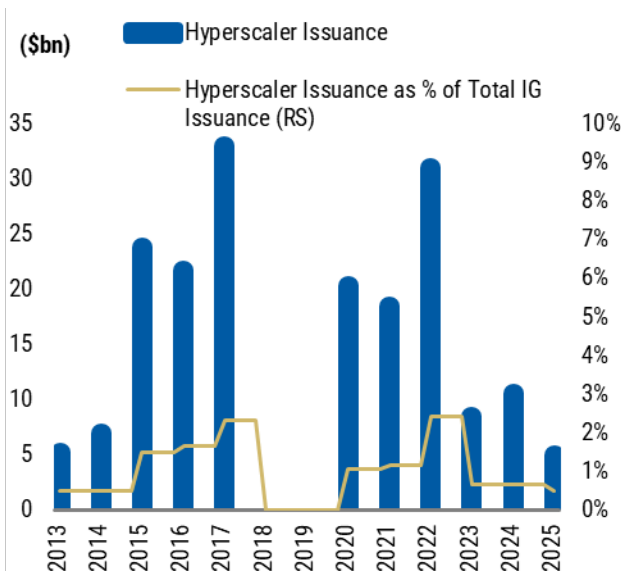
We expect \$200bn of corporate debt issuance

While hyperscalers would be able to issue >\$500bn before impacting their ratings, these companies may be averse to higher on-balance-sheet debt-funding for capex

Hyperscaler Debt Out. As % of Total IG Market



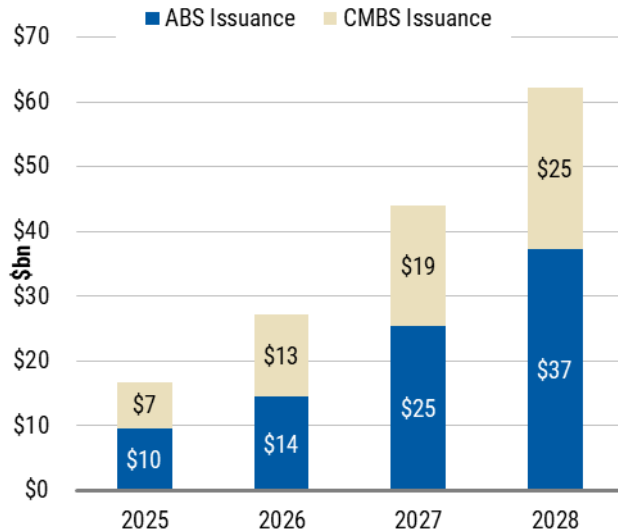
Hyperscaler \$ Bond Issuance



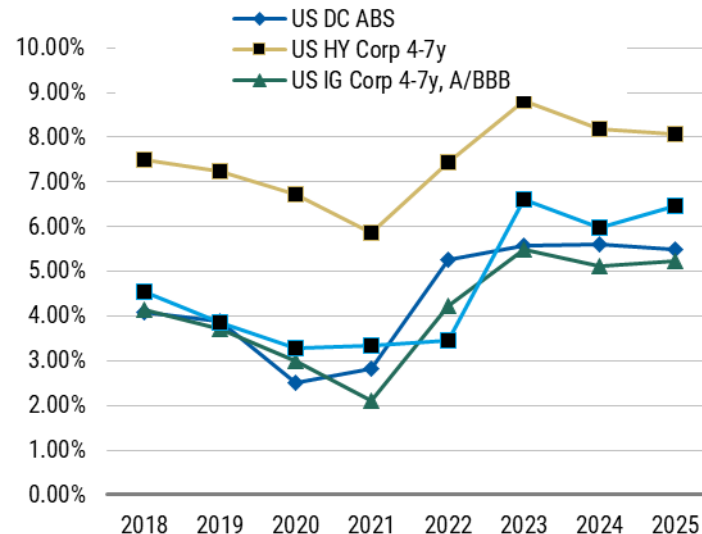
Securitized Credit Markets to finance \$150bn through 2028

We see an increasing rate of securitization, from 10% at present growing to 25% by 2028; developers achieve competitive financing costs through securitization

Est. Annual Issuance for DC Securitized Credit



Est. Financing Costs by Market



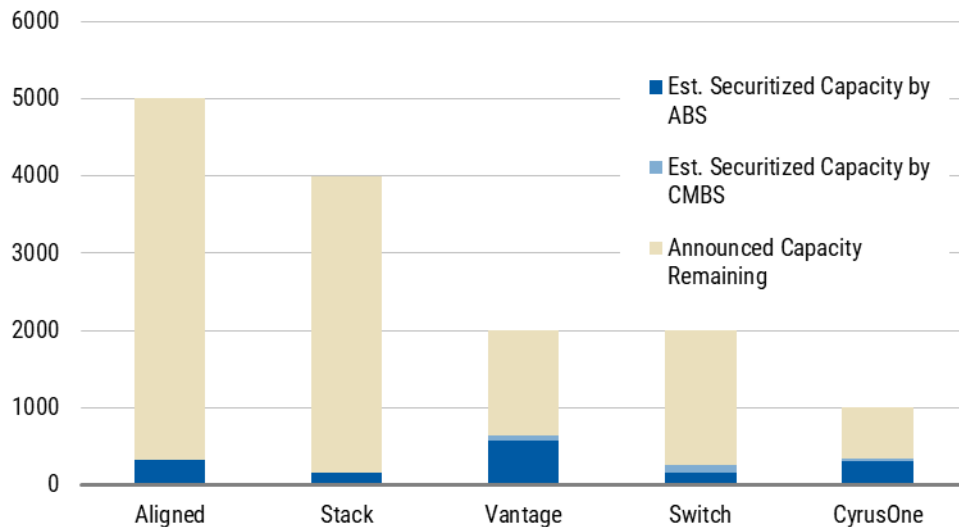
Securitization rate to climb from 10% today to 25% by 2028 (MS Est.)

Securitized credit issuers have ambitious global growth plans, but have to date securitized just 10% of their planned or operational capacity (by MWs).

Combined, the ABS issuers have >25GW of planned capacity but have securitized <3GW to date.

While 90% of securitized MWs are at present in the US, we see scope for greater expansion into Canadian and European markets as well.

MWs in Securitized Deals vs Global Capacity Ambition, Leading Issuers



An \$800bn opportunity for private credit, led by asset-based financing (ABF)

Total Addressable Market for ABF Significantly Larger than Direct Lending

We think private capital – in particular, credit – will play a key role in meeting a majority of the remaining financing gap as it sits optimally at the intersection of significant expansion in AUM in a higher-rate environment and the complex, global, and customized financing needs that are associated with AI build-out.



What are investors asking?

How will GPUs be financed, particularly for non-IG buyers?

While it is clear that hyperscalers have both the cash on hand and access to capital markets necessary to buy GPUs and build/lease data centers as needed, investors are more skeptical about non-IG and relatively new AI players' ability to finance their purchases of GPUs in size and over time. Loans backed by GPUs that amortize over time (and over a shorter period than the chips' useful life) is likely to be a popular solution. Moreover, companies that lease these GPUs to high-quality counterparties benefit from strong contractual terms that ensure cash flow over the term of the loan. We think public securitization could be a potential solution, but not likely until later this decade. HY unsecured issuance is also an option but will come at a cost to the borrower with rates in the high-single digits to mid-teens.

Is this another credit bubble? Why is this time different?

Seasoned credit investors will recall that through the later part of the 90s, corporates were very aggressive with their capital allocation, and debt/leverage increased sharply. The period ultimately ended with the sharpest default cycle in recent history from 2000-04. Clearly, debt growth of this magnitude runs into the risk of obsolescence, slower monetization, and weak macro conditions. An encouraging sign, though, is the diverse pools of capital that are available today, which can distribute the warehousing of credit risk (unlike the late 1990s when it was concentrated on corporate balance sheets). The ultra-high-quality credit profile of hyperscalers and significant cash on hand mean less sensitivity to macro conditions.

Morgan Stanley

Powering AI: Bottlenecks and Potential Solutions

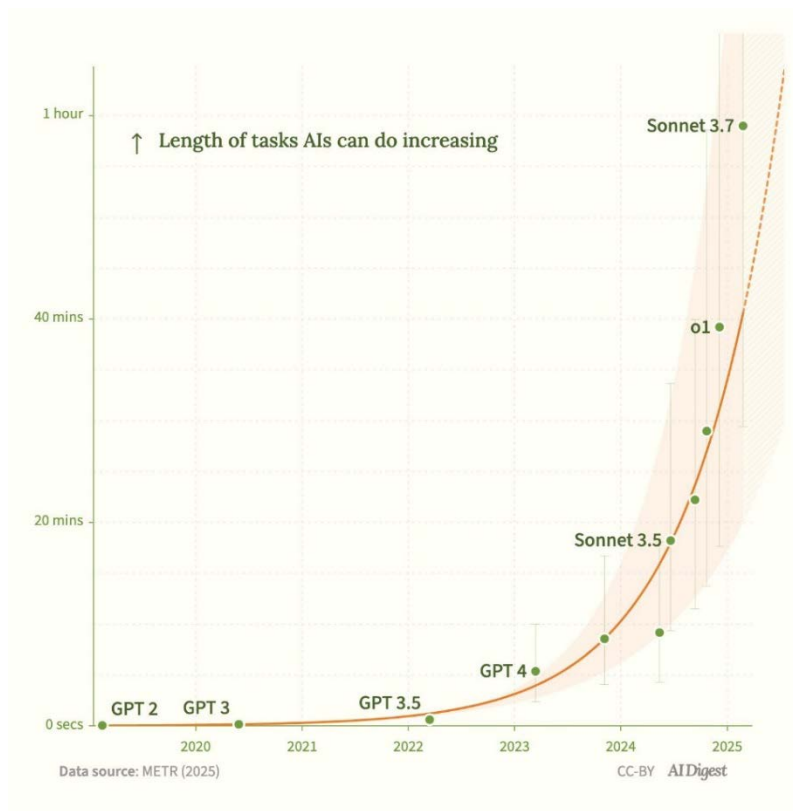
Stephen Byrd, Jamil Ohayia, David Arcaro

We Are Bullish Regarding the Non-Linear Rate of AI Capability Improvement

An Example: Agentic AI

From Model Evaluation & Threat Research, an independent AI assessment organization:

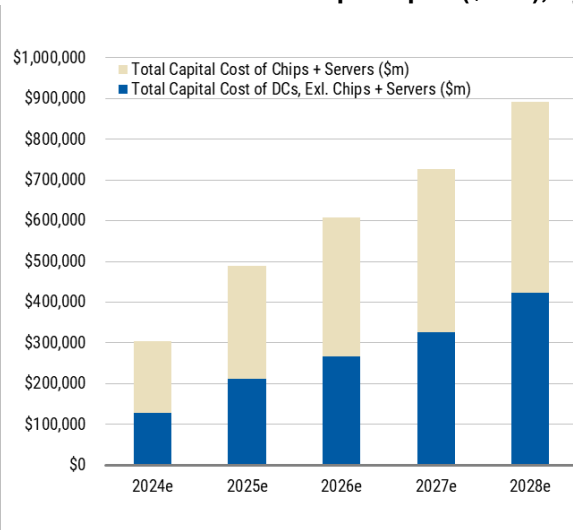
“We propose measuring AI performance in terms of the *length* of tasks AI agents can complete. We show that this metric has been consistently exponentially increasing over the past 6 years, with a doubling time of around 7 months. Extrapolating this trend predicts that, in under five years, we will see AI agents that can independently complete a large fraction of software tasks that currently take humans days or weeks.”



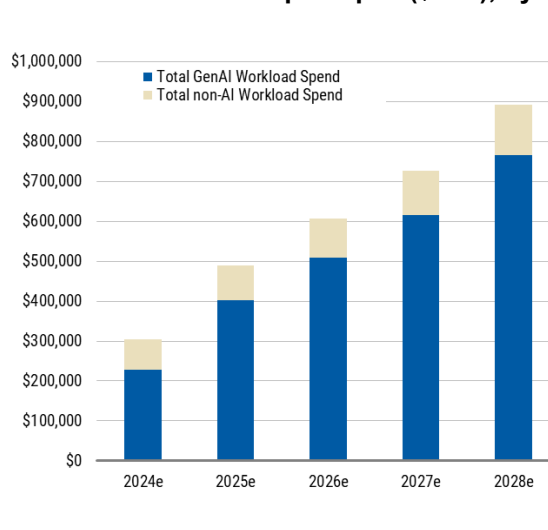
AI Infrastructure: Total Cumulative Spend of >\$3 Trillion Through 2028

- ~\$2.6 trillion for data centers, including chips and servers, through 2028
- >110 gigawatts (GW) of power needed through 2028; at \$2-3/watt, power plants would cost ~\$210-330b
- Power grid spend on AI is more difficult to isolate, but is likely in the hundreds of billions through 2028

Est. Total Global Data Center Spend p.a. (\$mm), by type



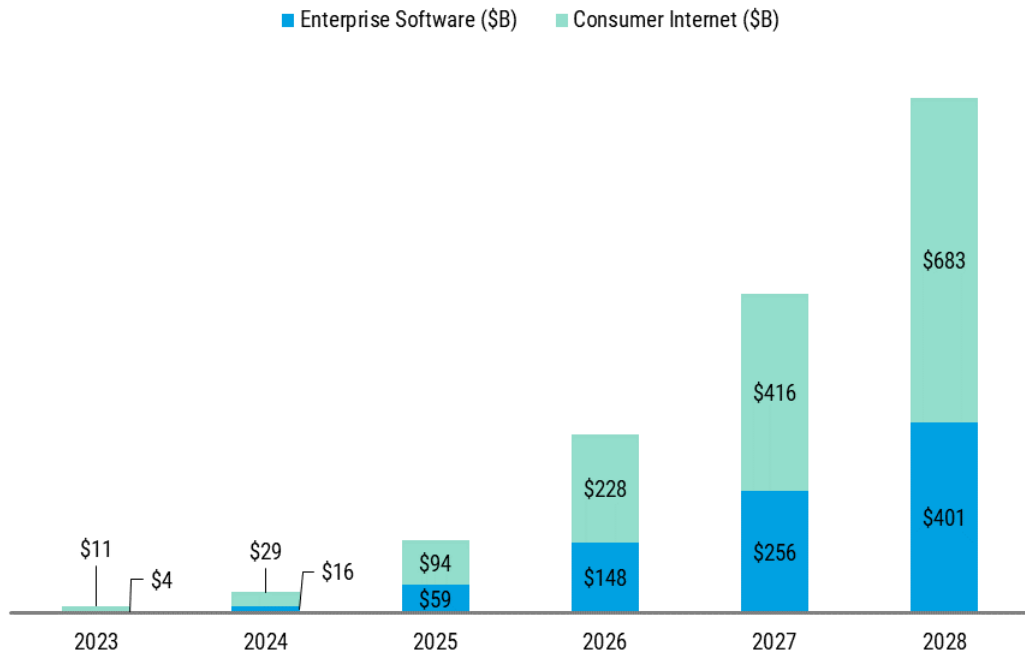
Est. Total Global Data Center Spend p.a. (\$mm), by workload



The “Return” Side of the GenAI ROI Equation

GenAI will drive a ~\$1 Trillion Revenue Opportunity by 2028

- We see **\$401B in GenAI Software spend by 2028**, from \$16B in 2024
- GenAI Software could represent ~22% of Total Software spending by 2028
- We see **\$683B in GenAI Consumer spending by 2028**, from \$29B in 2024
- GenAI Consumer spending is likely led by Ecommerce, Search, and Autonomous



AI Adoption Use Case: Coding

Microsoft: as much as 30% of code currently written by AI

Anthropic CEO: within 3-6 months, 90% of code will be produced by AI

Coding-related US jobs:

- Computer programmers: 140k jobs, \$100k avg salary
- Web developers: 223k jobs, \$96k avg salary
- Software developers: 1.7m jobs, \$131k avg salary

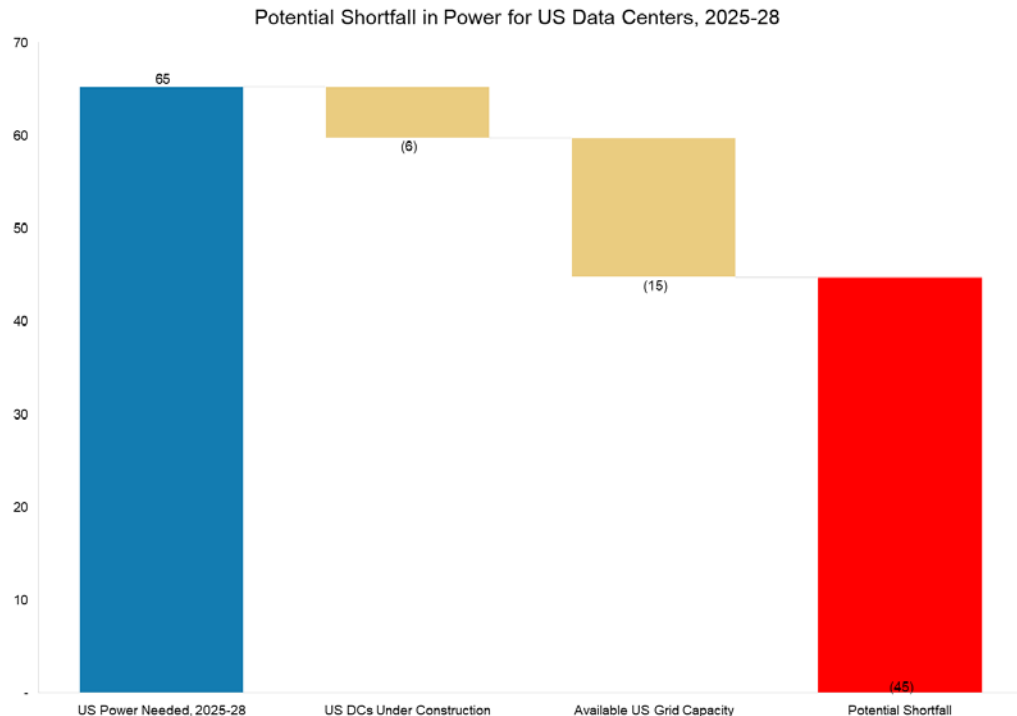
Model	Percent correct	Cost	Command	Correct edit format
o3 (high) + gpt-4.1	82.7%	\$69.29	<code>aider --model o3 --architect</code>	100.0%
o3 (high)	79.6%	\$111.03	<code>aider --model o3</code>	95.1%
Gemini 2.5 Pro Preview 05-06	76.9%	\$37.41	<code>aider --model gemini/gemini-2.5-pro-preview-05-06</code>	97.3%
Gemini 2.5 Pro Preview 03-25	72.9%		<code>aider --model gemini/gemini-2.5-pro-preview-03-25</code>	92.4%
o4-mini (high)	72.0%	\$19.64	<code>aider --model o4-mini</code>	90.7%
claude-3-7-sonnet-20250219 (32k thinking tokens)	64.9%	\$36.83	<code>aider --model anthropic/claude-3-7-sonnet-20250219 --thinking-tokens 32k</code>	97.8%
DeepSeek R1 + claude-3-5-sonnet-20241022	64.0%	\$13.29	<code>aider --architect --model r1 --editor-model sonnet</code>	100.0%

The US Power Bottleneck

Our [Powering AI model](#) is a helpful tool to understand the drivers of data center power demand growth – our power demand numbers are above consensus

We see multiple bottlenecks to data center growth in the US and Europe: (A) grid access, (B) power equipment, (C) labor, and (D) political capital

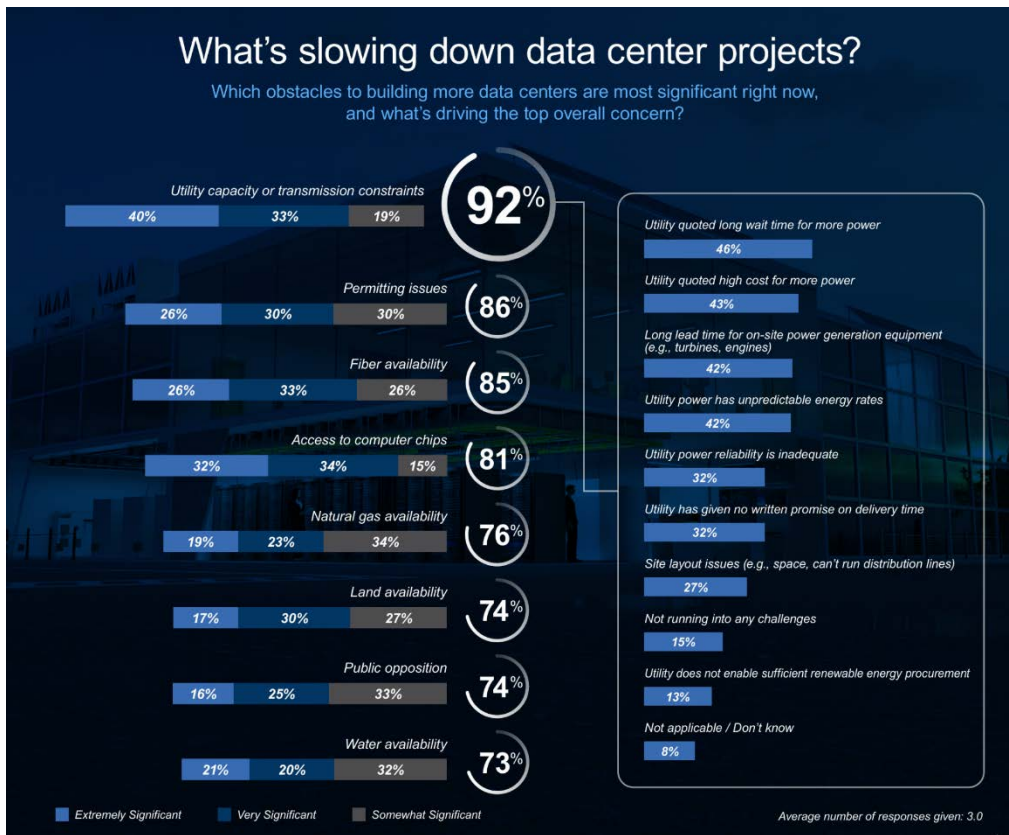
What are the best “time to power” solutions? (A) siting at nuclear power plants, (B) natural gas-fired plants and fuel cells, (C) conversion of crypto and industrial sites



A Recent Schneider Electric Survey Shows Power Issues are Paramount

Key survey takeaways:

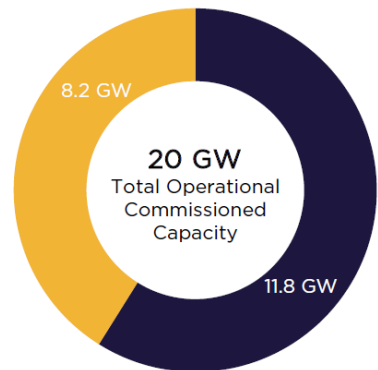
1. Grid constraints are the #1 barrier...
2. New projects are bigger than ever: Nearly half (48%) of respondents say their new data centers now average 100+ MW, signaling an unprecedented scale of development.
3. Longer lead times are the new normal: 44% report utility wait times of 4+ years, while demand is spilling into secondary markets, stretching available capacity.
4. The AI boom is impacting sustainability goals...
5. Alternatives are emerging: 62% of respondents reported exploring on-site power generation. With no slowdown in sight, data center operators are rethinking power procurement strategies”



What Is the Compute Used for? 75%/25% Inference/Training in the US

- Cushman & Wakefield is tracking 47 gigawatts of US data centers in development; 4 GW is “hyperscale”
- Our 62 gigawatts of US data center demand through 2028, net of ~44 gigawatts of Inference-focused data centers, would imply ~18 gigawatts of Training-focused data centers (~1/4 of our 62 GW total)
- This resulting ~18 gigawatts for Training through 2028 makes sense, given we have several US companies that each plan to have multiple gigawatts for Training

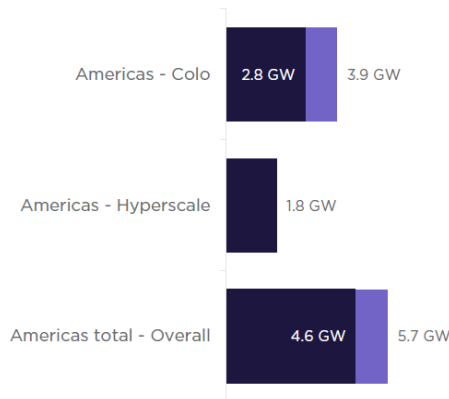
**Americas Operational
Commissioned Capacity (Including
Expansions)**



■ Americas - Colo ■ Americas - Hyperscale

Source: datacenterHawk, DC Byte, Cushman & Wakefield Research

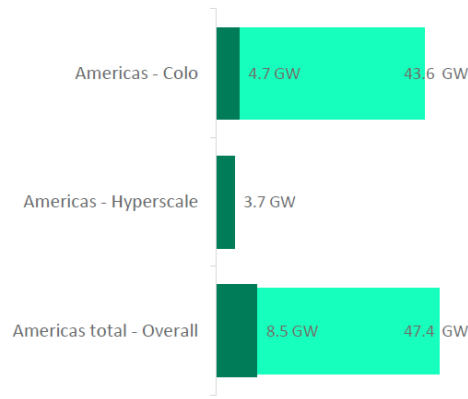
**Landscape of Americas Data
Centers Under Construction**



■ Total Under Construction (MW)
■ Under Construction Commissioned Capacity

Source: datacenterHawk, DC Byte, Cushman & Wakefield Research

**Landscape of Americas Planned
Data Centers**



■ Total Planned (MW)
■ Planned Commissioned Capacity

Source: datacenterHawk, DC Byte, Cushman & Wakefield Research

The Most Attractive Grid “De-Bottlenecking” Solutions

- **Definition of a “de-bottlenecking” solution:** any product that can result in data centers powering up more rapidly than the default approach (file a grid interconnection request at a greenfield site)
- **Option #1: build power plants at the data center site, and go off-grid while waiting for grid connection approval**
 - Favors natural gas and coal power, but we could also see renewables, nuclear power and energy storage
 - Significant potential operational issues with not being connected to the grid
- **Option #2: build the data center at a power plant or industrial site with ample power infrastructure**
 - Increasingly facing political and grid operator pushback; tends to result in higher power prices
- **Option #3: redirect power available to a Bitcoin site or industrial power user**
 - We view this as an attractive option, but not large enough to resolve the bottleneck issue
- **Our concern: many of these options face significant execution risk - the risk of a bottleneck is high**

Powering AI: Why “Time to Power” Has Been a Powerful Alpha Driver

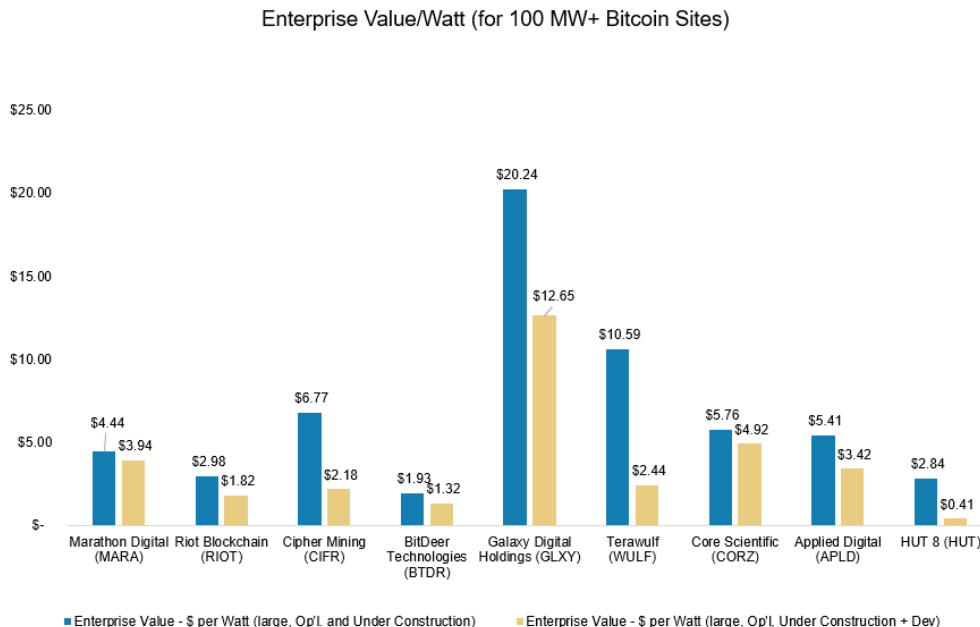
- **Definition of a “time to power” solution: any product that can result in data centers powering up more rapidly than the default approach (file a grid interconnection request)**
 - Many US grids now have a significant backlog of requests, and the typical grid connection time for data centers has increased from ~2-3 years to ~5-10 years
- **Powering AI Alpha is driven by a fascinating collision of two worlds**
 - AI data centers: facility cost of **~\$30m per megawatt**, with rapidly depreciating chips
 - A power plant often costs **~\$2-3 million per megawatt**, with a very long asset life
- Our “time to power” math suggests that the value of a 1-year time advantage is \$3-4m per megawatt, more than the entire cost of most power plants
- **The “de-bottlenecking” transactions we have seen appear to include large premia from the lens of the power world, but the premia appear relatively small from the lens of an AI player**
- Bitcoin conversion: a few transactions at effective power price premia of >300%

Bitcoin Stocks Trade at Low Enterprise Value/Watt Levels

Many Bitcoin stocks trade at an EV/watt of \$1-2, well below likely value of converting large sites to HPC data centers – quite low given the CoreWeave annual lease payments to CORZ are ~\$1.50/watt *per year* for 12 years

We would expect to continue to see conversion transactions at relatively high implied EV/watt levels

[Please ask us for a copy of our most recent model that shows this math](#)



Potential HPC Data Center Construction Financing Structures

WULF Compute – Potential Illustrative Business Models

Maximizing land, megawatts, and capital for optimal long-term value creation

WULF Compute's Core Business

	Cloud Service Provider (CSP)	Colocation – Whitespace & Rack Ready	Build to Suit
Description	Shell data center and GPU clusters; space, power, cooling, fiber/internet, security, compute, storage and environment.	Shell data center; lease rack ready whitespace including redundant power, cooling, fiber/internet, and security	Data center built to spec; lease space, redundant power, cooling, fiber/internet, and security
Customer	Multiple – Managed through cloud partner	Multiple – Enterprise and well funded startups	Single – Hyperscaler
Contract Size	0.1 – 50 MW	0.5 – 100 MW+	100+ MW
Contract Term	1 – 36 months	5 – 15 years	15+ years with renewals
O&M	WULF managed	WULF managed	Tenant managed
Build Cost per MW	\$27 – 30 million ⁽¹⁾	\$6 – 8 million ⁽¹⁾	\$2 – 4 million ⁽⁴⁾
Financing	Equity & Debt	Equity (initially)	Highly Bankable
Revenue per MW	\$11 – 18 million ⁽²⁾	\$1.3 – 1.8 million ⁽³⁾	\$0.2 – 0.5 million
Margins	70 – 80%	65 – 75%	90% +
Valuation Range	7-12x EBITDA	10-15x EBITDA	15-20x EBITDA



(1) Build cost for CSP and colocation model reflect liquid cooling infrastructure. Low end of range based on long-term reserve contract at \$2.1/GPU/hr with 100% rental utilization, high end of range represents \$4.50/GPU/hr of on-demand revenue at 80% rental utilization.

(3) Does not include customer pass-through energy expense. Subject to term length and payment terms.

(4) Based on air cooled powered shell infrastructure. NNN Rent Structure; customer pays all OpEx directly.

The “REIT End Game” for Bitcoin Assets Converted to HPC Data Centers

	Range					
Data Center Total Power (MW)		100		100		100
Data Center Revenue/Watt	\$	1.55	\$	1.55	\$	1.55
Total Data Center Revenue (\$m per year)		\$155		\$155		\$155
Operating Margin		70%		70%		70%
Operating Costs	\$	(47)	\$	(47)	\$	(47)
EBITDA (\$m)		\$109		\$109		\$109
Total Construction Cost/Watt		\$6		\$7		\$8
Construction Cost (\$m)	\$	600	\$	700	\$	800
Construction Costs as a Multiple of EBITDA		5.5		6.5		7.4
Leverage as a Multiple of EBITDA		4.0		4.0		4.0
Leverage (% of Cost)		72%		62%		54%
Indicative Enterprise Value/EBITDA Multiple		10.0x		12.5x		15.0x
Resulting Enterprise Value (\$m)	\$	1,085	\$	1,356	\$	1,628
Resulting Equity Value (\$m)	\$	651	\$	922	\$	1,194
Value Creation on Equity Investment		292%		247%		226%
Equity Value (\$m per megawatt)	\$	6.51	\$	9.22	\$	11.94
Equity Value Creation (\$m per megawatt)	\$	4.85	\$	6.56	\$	8.28

Based on our indicative assumptions, returns could be high for building and leasing a “powered shell” to a hyperscaler, followed by selling this asset to a REIT – these value creation levels are well above where most Bitcoin stocks trade on an EV/watt basis

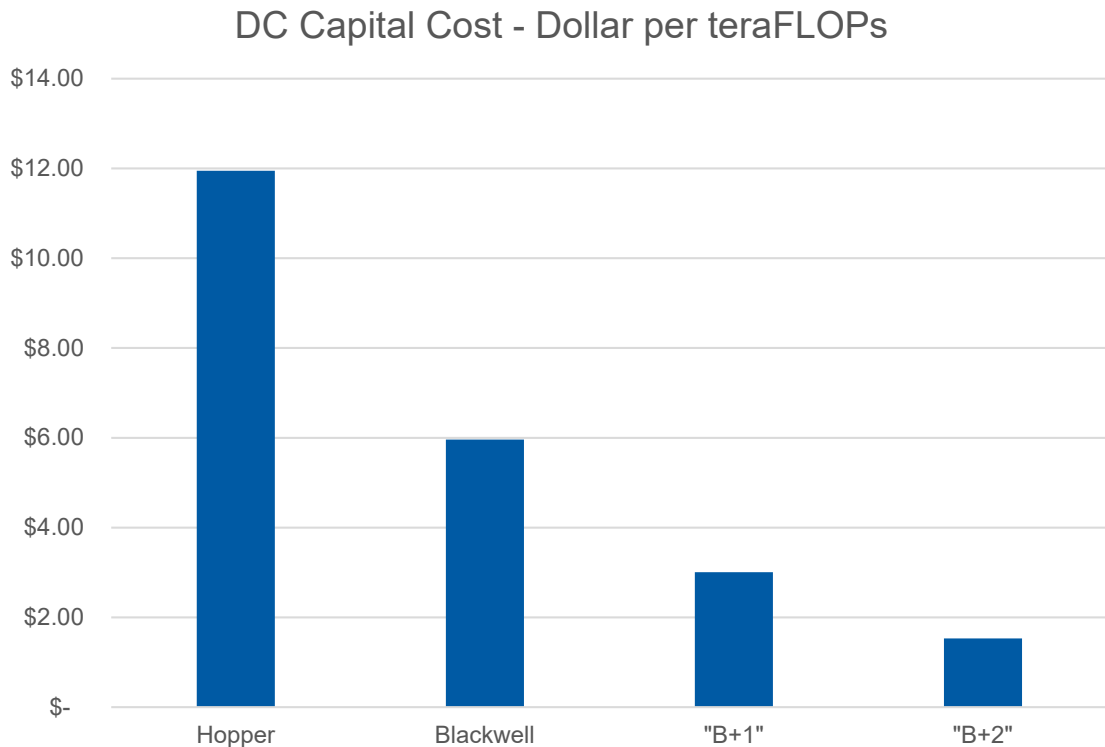
Is DeepSeek a Bearish Signal for Data Center Demand?

We believe the answer is no

1. More innovation drives more complex models, and consequently more demand for compute
2. The level of AI adoption is under-appreciated
3. The value of non-linear AI improvement is under-appreciated - a quote from Dario Amodei, CEO of Anthropic: *“all else equal, scaling up the training of AI systems leads to smoothly better results on a range of cognitive tasks, across the board...”* These differences tend to have huge implications in practice — another factor of 10 may correspond to the difference between an undergraduate and PhD skill level — and thus companies are investing heavily in training these models...The field is constantly coming up with ideas, large and small, that make things more effective or efficient: it could be an improvement to the *architecture* of the model...or simply a way of running the model more efficiently on the underlying hardware. New generations of hardware also have the same effect. What this typically does is [shift the curve](#): if the innovation is a 2x ‘compute multiplier’ (CM), then it allows you to get 40% on a coding task for \$5M instead of \$10M; or 60% for \$50M instead of \$100M, etc...**Because the value of having a more intelligent system is so high, this shifting of the curve typically causes companies [to spend more](#), not less, on training models: the gains in cost efficiency end up entirely devoted to training smarter models, limited only by the company's financial resources.** People are naturally attracted to the idea that ‘first something is expensive, then it gets cheaper’ — as if AI is a single thing of constant quality, and when it gets cheaper, we'll use fewer chips to train it. But what's important is the *scaling curve*: when it shifts, we simply traverse it faster, because the value of what's at the end of the curve is so high.”

GPU Replacement Cycle – Innovation Drives Depreciation Risk

The cost per unit of computational power (technically, capital cost divided by teraFLOPs) has been dropping rapidly; over a 6-year period, we would expect the capital cost per unit of computational power to drop by ~90%



Data Center Operational Risk: Grid Instability

This lightning event raised alarm bells among US grid regulators – a loss of a transmission line caused voltage instability, and a very large number of data centers tripped offline as a result



NERC
NORTH AMERICAN ELECTRIC
RELIABILITY CORPORATION

Data Center Load Loss Event



Incident Review
Considering Simultaneous Voltage-Sensitive Load Reductions

Primary Takeaways
Operators and planners of the Bulk Electric System (BES) should be aware of the risks and challenges associated with voltage-sensitive large loads that are rapidly being connected to the power system. Specifically, when considering data centers and cryptocurrency mining facilities, entities should be aware of the potential for large amounts of voltage-sensitive load loss during normally cleared faults on the BES. Voltage-sensitive data center-type loads have increased on the system and are predicted to continue growing rapidly. The 2024 NERC Long-Term Reliability Assessment (LTRA) documents and discusses this potential growth of data center-type loads. This vignette highlights this load-loss potential based on analysis of a recent event in the Eastern Interconnection and offers some considerations for BES operators, planners, and regulators concerning identifying and mitigating the potential reliability effects and risks presented by these large voltage-sensitive load losses for future operations.

Summary of Incident
A 230 kV transmission line fault led to customer-initiated simultaneous loss of approximately 1,500 MW of voltage-sensitive load that was not anticipated by the BES operators. The electric grid has not historically experienced simultaneous load losses of this magnitude in response to a fault on the system, which has historically been planned for large generation losses but not for such significant simultaneous load losses. Simultaneous large load losses have two effects on the electric system: first, frequency rises on the system as a result of the imbalance between load and generators; second, voltage rises rapidly because less power is flowing through the system. In this incident, the frequency did not rise to a level high enough to cause concern. The voltage also did not rise to levels that posed a reliability risk, but operators did have to take action to reduce the voltage to within normal operating levels. However, as the potential for this type of load loss increases, the risk for frequency and voltage issues also increases. Operators and planners should be aware of this reliability risk and ensure that these load losses do not reach intolerable levels.

Incident Details
At approximately 7:30 p.m. Eastern on July 10, 2024, a lightning arrester failed on a 230 kV transmission line in the Eastern Interconnection, resulting in a permanent fault that eventually "locked out" the transmission line. The auto-reclosing control on the transmission line was configured for three auto-reclose attempts (staggered at each end of the line). This configuration resulted in 6 successive system faults in an 82-second period. The protection system detected these faults and cleared them properly. The shortest fault duration was the initial fault at 42 milliseconds, and the longest fault duration was 66 milliseconds. The voltage magnitudes during the fault ranged from .25 to .40 per unit in the load-loss area.

Event:

- 1,500MW load loss, exclusively data center load
- Coincident with 230kV normal line fault clearing
- Widespread: 60 different load points, 25 substations

Conclusions:

- Require models for large loads to enable studies to determine risk to BES for coincident large load losses
- Determine if large loads should be a NERC Registered Entity
- Need for new or modifications to Standards

4

RELIABILITY | RESILIENCE | SECURITY

A Key “Multipolar World” Dynamic: Possible AI Tech Restrictions

We believe there is a meaningful probability that the Trump administration will take a more expansive approach to AI technology restrictions on China – a few data points:

- **Dario Amodei, CEO of Anthropic:** *“We shouldn’t hand the Chinese Communist Party technological advantages when we don’t have to...Making AI that is smarter than almost all humans at almost all things will require millions of chips, tens of billions of dollars (at least), and is most likely to happen in 2026-2027...If China can’t get millions of chips, we’ll (at least temporarily) live in a unipolar world, where only the US and its allies have these models...[I]n this world, the US and its allies might take a commanding and long-lasting lead on the global stage.”*
- **President Trump:** *DeepSeek “should be a wake-up call for our industries that we need to be laser focused on competing”*
- **US House of Representatives Homeland Security Chairman Mark Green (R-Tenn.):** *“My proposed legislation will establish safeguards to prevent future shocks like China’s development of DeepSeek using American technology. In addition to the chips China reportedly stockpiled, it appears China used chips under the current export control threshold to achieve this AI breakthrough”*
- **Jacob Helberg, research fellow at the Palantir Foundation and a Commissioner for the U.S.-China Economic & Security Review Commission:** *“AI is the essential omni-purpose technology of the U.S.-China arms race, like steel in medieval warfare”*

Morgan Stanley

AI Adoption: Key Takeaways From Our Mapping Exercise

Todd Castagno

Key Takeaways from Our 4th AI Mapping Exercise

- **Broadening of AI exposure and materiality, reaching multi-trillion-dollar scale:** 412 stocks (\$8.7 trillion of market cap) have increased their exposure to AI and it is more material to 259 stocks (\$8.5 trillion of market cap). We count 102 stocks (\$2.4 trillion of market cap) that increased in both exposure to and materiality of AI
- **There are clear signs of alpha from AI materiality among companies: this can be seen in relative price performance and earnings revisions**
- **This Round, the Big AI "rate of change" winners are the Consumer and Real Estate sectors:** Almost 30% of Consumer Durables & Apparel stocks saw an increase in exposure, and ~10% experienced an increase in materiality. The percent of Consumer Durables & Apparel companies now qualifying as Adopters moved from 20% to 44%. Consumer Discretionary Adopters jumped from 32% to 47%. 32% of REITs are now more exposed to AI
- **Financials see significant increases in exposure:** Insurance Adopters increased from 48% to 71%, while Financial Services Adopters increased from 66% to 73%
- **Regionally, Asia Pacific is the most notable leader:** AI exposure increased by 104 companies (net), and 30 of these stocks also saw an increase in materiality

AI Adoption: Thematic Investment Strategies

- **Strategy #1: Overweight-rated stocks with both materiality and exposure increases:** we favor this approach; (A) it focuses on stocks where Morgan Stanley analysts see attractive risk-reward and (B) where stocks are experiencing "AI rate of change," which is a powerful alpha signal. >\$30 billion market cap stocks in this category include **S&P Global, EssilorLuxottica, Siemens Energy, Ecolab, Zijin Mining Group, Johnson Controls International, and Titan Company**
- **Strategy #2: New "Core to Thesis" stocks:** **Amazon.com, SAP, IBM, Johnson Controls International, and ASM International** are >\$30 billion market cap companies on this list
- **Strategy #3: Adopters with pricing power:** These stocks have been breaking away from the market, and we believe that pricing power will continue to be a key driver of alpha going forward, determining the extent to which AI-driven benefits and gains will be retained. Large companies identified as Adopters with strong pricing power include companies such as **Microsoft, Marriott International, DoorDash, Rolls-Royce, and Home Depot**
- **Strategy #4: Stocks with the greatest level of AI materiality and pricing power:** these stocks tend to be central to AI proliferation, and their involvement ranges from key Enablers, Adopters, and Infrastructure plays. This strategy represents a relatively broad approach to investing in this theme. **NVIDIA, Microsoft, Apple, Amazon.com, Meta Platforms, Alphabet, Broadcom, and TSMC** are the largest stocks in this bucket

Data Center Operational Risk: Grid Instability

This lightning event raised alarm bells among US grid regulators – a loss of a transmission line caused voltage instability, and a very large number of data centers tripped offline as a result



NERC
NORTH AMERICAN ELECTRIC
RELIABILITY CORPORATION

Data Center Load Loss Event



Incident Review
Considering Simultaneous Voltage-Sensitive Load Reductions

Primary Takeaways
Operators and planners of the Bulk Electric System (BES) should be aware of the risks and challenges associated with voltage-sensitive large loads that are rapidly being connected to the power system. Specifically, when considering data centers and cryptocurrency mining facilities, entities should be aware of the potential for large amounts of voltage-sensitive load loss during normally cleared faults on the BES. Voltage-sensitive data center-type loads have increased on the system and are predicted to continue growing rapidly. The 2024 NERC Long-Term Reliability Assessment (LTRA) documents and discusses this potential growth of data center-type loads. This vignette highlights this load-loss potential based on analysis of a recent event in the Eastern Interconnection and offers some considerations for BES operators, planners, and regulators concerning identifying and mitigating the potential reliability effects and risks presented by these large voltage-sensitive load losses for future operations.

Summary of Incident
A 230 kV transmission line fault led to customer-initiated simultaneous loss of approximately 1,500 MW of voltage-sensitive load that was not anticipated by the BES operators. The electric grid has not historically experienced simultaneous load losses of this magnitude in response to a fault on the system, which has historically been planned for large generation losses but not for such significant simultaneous load losses. Simultaneous large load losses have two effects on the electric system: first, frequency rises on the system as a result of the imbalance between load and generators; second, voltage rises rapidly because less power is flowing through the system. In this incident, the frequency did not rise to a level high enough to cause concern. The voltage also did not rise to levels that posed a reliability risk, but operators did have to take action to reduce the voltage to within normal operating levels. However, as the potential for this type of load loss increases, the risk for frequency and voltage issues also increases. Operators and planners should be aware of this reliability risk and ensure that these load losses do not reach intolerable levels.

Incident Details
At approximately 7:30 p.m. Eastern on July 10, 2024, a lightning arrester failed on a 230 kV transmission line in the Eastern Interconnection, resulting in a permanent fault that eventually "locked out" the transmission line. The auto-reclosing control on the transmission line was configured for three auto-reclose attempts staggered at each end of the line. This configuration resulted in 6 successive system faults in an 82-second period. The protection system detected these faults and cleared them properly. The shortest fault duration was the initial fault at 42 milliseconds, and the longest fault duration was 66 milliseconds. The voltage magnitudes during the fault ranged from .25 to .40 per unit in the load-loss area.

Event:

- 1,500MW load loss, exclusively data center load
- Coincident with 230kV normal line fault clearing
- Widespread: 60 different load points, 25 substations

Conclusions:

- Require models for large loads to enable studies to determine risk to BES for coincident large load losses
- Determine if large loads should be a NERC Registered Entity
- Need for new or modifications to Standards

4

RELIABILITY | RESILIENCE | SECURITY

Hyperscaler cash flow and tax benefits from the OBBBA

Amazon: Largest Beneficiary with a ~30% (~\$15bn) Lift to '26 FCF

Meta: Potential ~\$8-10bn Tailwind to META FCF in Medium Term

Google: Biggest Near-Term Beneficiary with \$25bn Tailwind in 2025, With Long-Term Tailwinds of \$4-\$6bn; beyond 2025, we see a \$4bn-\$6bn benefit to GOOGL (~5-6% uplift to FCF)

	2025E	2026E	2027E	2028E
AMZN				
Current FCF	30,209	48,046	61,519	83,820
OBBBA Impact	<u>14,873</u>	<u>14,636</u>	<u>14,539</u>	<u>11,325</u>
FCF after OBBBA	45,083	62,682	76,058	95,146
% change	49%	30%	24%	14%
GOOGL				
Current FCF	78,821	85,520	95,658	108,517
OBBBA Impact	<u>24,724</u>	<u>4,170</u>	<u>6,047</u>	<u>5,372</u>
FCF after OBBBA	103,545	89,690	101,705	113,888
% change	31%	5%	6%	5%
META				
Current FCF	44,872	47,429	56,538	70,439
OBBBA Impact	<u>9,893</u>	<u>10,351</u>	<u>10,149</u>	<u>8,140</u>
FCF after OBBBA	54,765	57,780	66,687	78,579
% change	22%	22%	18%	12%

Source: Morgan Stanley Research estimates

Disclosures

The information and opinions in Morgan Stanley Research were prepared by Morgan Stanley & Co. LLC, and/or Morgan Stanley C.T.V.M. S.A., and/or Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., and/or Morgan Stanley Canada Limited. As used in this disclosure section, "Morgan Stanley" includes Morgan Stanley & Co. LLC, Morgan Stanley C.T.V.M. S.A., Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., Morgan Stanley Canada Limited and their affiliates as necessary.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/researchdisclosures, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: David Arcaro, CFA; Stephen C Byrd; Carolyn L Campbell; Jamil Ohayia; Vishwas Patkar; Lindsay A Tyler.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of June 30, 2025)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent

Disclosures (cont.)

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1497	40%	372	45%	25%	702	41%
Equal-weight/Hold	1632	44%	380	46%	23%	776	45%
Not-Rated/Hold	4	0%	0	0%	0%	2	0%
Underweight/Sell	593	16%	79	10%	13%	233	14%
Total	3,726		831			1713	

recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months. Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Disclosures (cont.)

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding the relationship between the companies that are the subject of Morgan Stanley Research and Morgan Stanley Smith Barney LLC or Morgan Stanley or any of their affiliates, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to www.morganstanley.com/researchdisclosures.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.

Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Disclosures (cont.)

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118J); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altium, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All

Disclosures (cont.)

recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

The following authors are neither Equity Research Analysts/Strategists nor Fixed Income Research Analysts/Strategists and are not opining on or expressing recommendations on equity or fixed income securities: Todd Castagno, CFA, CPA.

The following authors are Fixed Income Research Analysts/Strategists and are not opining on or expressing recommendations on equity securities: Carolyn L Campbell; Vishwas Patkar; Lindsay A Tyler.

The Americas

1585 Broadway
New York, NY 10036-8293
United States
+1 212 761 4000

Europe

20 Bank Street, Canary Wharf
London E14 4AD
United Kingdom
+44 (0)20 7425 8000

Japan

1-9-7 Otemachi, Chiyoda-ku
Tokyo 100-8104
Japan
+81 (0) 3 6836 5000

Asia/Pacific

1 Austin Road West
Kowloon
Hong Kong
+852 2848 5200